FULL SIMULATION SET 02

UPSC Civil Services (Main) Simulation GS-III

DETAILED SOLUTIONS / ANSWER KEY | Time: 3 Hours | Maximum Marks: 250

Q1. Why do the three methods of measuring GDP converge conceptually yet diverge statistically?

[10 marks | 150 words]

Model answer (148 words; practice model, not official): The production, income and expenditure methods observe the same circular flow. Production creates value added; that value becomes compensation, operating surplus and mixed income; final users absorb the resulting output through consumption, capital formation and net exports. Conceptually, therefore, output equals income equals final expenditure. Measured totals differ because the routes use distinct evidence. Production estimates combine enterprise surveys, crop and industrial indicators and administrative records. Income data face mixed household-enterprise accounts and under-reporting. Expenditure estimates depend on consumption, inventory, trade and government data with different reporting dates. Price adjustment, informal-sector extrapolation and later returns add further variation. MoSPI's 27 February 2026 methodology uses Supply and Use Table integration to improve production- expenditure reconciliation. The residual remains a statistical discrepancy, not additional production. Better surveys, coherent classifications and transparent revision tables can reduce it, but independent estimation means a zero discrepancy is not a realistic proof of quality.

Q2. How did colonial land-revenue systems shape the agenda of post-Independence land reform?

[10 marks | 150 words]

Model answer (122 words; practice model, not official): Colonial revenue settlements created distinct obstacles. The Permanent Settlement of 1793 placed zamindars between the State and cultivators, strengthening rent-receiving interests. Ryotwari dealt directly with recorded cultivators, but direct assessment did not guarantee affordable demand or secure title. Mahalwari made a village estate or co-sharer body responsible, tying revenue administration to local hierarchy. Post-Independence reform therefore required several instruments. Intermediary abolition removed rent-receiving layers; tenancy laws regulated rent, eviction and ownership; ceilings addressed concentration; consolidation tackled scattered parcels; and records identified actual rights. The causal lesson is that a revenue label did not determine one outcome. Regional variants, political power and record quality shaped each legacy. Reform had to alter both legal relationships and the administrative information through which they were enforced.

Q3. Explain the principle of optimal risk allocation in infrastructure PPPs.

[10 marks | 150 words]

Model answer (121 words; practice model, not official): Risk allocation determines whether a PPP creates value. Each risk should rest with the party able to control its probability or mitigate its impact at lowest cost. Private parties may manage construction; public authorities manage land and sovereign clearances. BOT Toll places traffic risk privately; annuity and HAM shift it toward government. Maximum private transfer is counterproductive. Unmanageable risks return as higher bids, financing cost, default or opaque renegotiation. Equally, excessive public guarantees convert a PPP into hidden borrowing. Contracts should identify cause-specific risks, provide measurable standards, disclose contingent liabilities and define relief, renegotiation and termination. Independent appraisal must compare retained public risk and lifecycle value with EPC. The goal is not to shift risk politically but to manage it economically.

Q4. Why did the 1991 industrial reforms not mean the end of the State?

[10 marks | 150 words]

Model answer (125 words; practice model, not official): The New Industrial Policy of 24 July 1991 removed most industrial licensing, reduced public-sector reservation, relaxed MRTP pre-entry controls and widened foreign-investment and technology routes. Allocation shifted from administrative permission toward firms and markets. The State's role changed rather than disappeared. Delicensed firms still require safety, environmental, land and sector approvals. Competitive markets need CCI oversight of agreements, abuse and combinations. Government provides infrastructure, skills, finance, standards and strategic capabilities and retains enterprises for sovereign, network or developmental reasons. The 2021 PSE Policy seeks bare-minimum public commercial presence in strategic sectors, not withdrawal from regulation or public goods. Thus 1991 replaced pervasive entry control with rule-based regulation and capability building. Effective markets still depend on capable public institutions. State capacity and accountability remain the test.

Q5. Explain why fiscal, revenue and primary deficits must be read together.

[10 marks | 150 words]

Model answer (146 words; practice model, not official): Fiscal deficit equals total expenditure minus non-debt receipts and broadly measures the year's borrowing requirement. Revenue deficit equals revenue expenditure minus revenue receipts and reveals current-account dissaving. Effective revenue deficit deducts grants for capital-asset creation, recognising that an item booked as Union revenue expenditure may create an asset elsewhere. Primary deficit subtracts interest payments from fiscal deficit and isolates the current non-interest gap. Together they answer different questions. A falling fiscal deficit may still conceal weak revenue balance; a high fiscal deficit may finance productive capital expenditure; and a low primary deficit may reflect a large inherited interest burden rather than low debt. Analysis must add financing mix, debt maturity, off-budget liabilities and expenditure outcomes. It must also preserve fiscal year and vintage. Union Budget 2026-27 figures are Budget Estimates, not realised outcomes. Therefore no single deficit is a complete measure of fiscal quality or sustainability.

Q6. Distinguish pool, flux and residence time in biogeochemical analysis.

[10 marks | 150 words]

Model answer (101 words; practice model, not official): 1. A biogeochemical cycle is movement of an element or compound between organisms and physical reservoirs; the analysis must identify the reservoir, the transfer process and the boundary within which a balance is claimed. 2. A pool is the quantity held in a reservoir and a flux is transfer per unit time; residence time requires a stated pool and outgoing flux, so none of the three may be replaced by an unsupported nutrient figure. 3. Photosynthesis transfers carbon into biomass, while respiration, decomposition and combustion return carbon to environmental reservoirs; a carbon stock and an annual carbon flux are different quantities.

Q7. Distinguish the roles of DoS, ISRO, IN-SPACe and NSIL.

[10 marks | 150 words]

Model answer (137 words; practice model, not official): 1. The Department of Space is the Union-government policy umbrella, while ISRO is the public space agency that develops launch vehicles, spacecraft, applications and missions; the department and agency are not interchangeable. 2. VSSC leads launch-vehicle development, liquid-propulsion work is distributed across LPSC and IPRC, and SDSC-SHAR integrates and conducts launches; naming a centre does not transfer the mandate of the whole agency to it. 3. IN-SPACe is the single-window nodal agency under the Department of Space for promoting, authorising and supervising non-government space activity; it is not ISRO's commercial sales arm. 4. NSIL is the public-sector commercial arm for launch, satellite, service and technology-transfer arrangements; commercial contracting is distinct from authorisation and supervision. 5. Indian Space Policy 2023 allocates institutional roles and opens participation, but it is executive policy rather than a comprehensive domestic space statute.

Q8. Explain the utility of the threat-vulnerability-capability framework for internal-security policy.

[10 marks | 150 words]

Model answer (109 words; practice model, not official): 1. Governance deficit, poverty, unemployment, inequitable growth, communal or caste tension, porous borders, hostile neighbours, corruption and a weak criminal-justice system are enabling conditions rather than interchangeable threats. 2. A threat is a hostile actor or event, a vulnerability is an exploitable weakness, capability is the State's actual preventive or responsive means, and consequence is the realised harm when the first exploits the second. 3. Prevention reduces vulnerability through intelligence, community trust and governance, whereas response contains an event; visible response outputs cannot substitute for harder-to-measure prevention. 4. A legal power is not operational capability: trained personnel, forensics, court time, lawful procedure and inter-agency trust remain separate implementation conditions.

Q9. Distinguish NDMA, NEC, SDMA, SEC and DDMA by composition and mandate.

[10 marks | 150 words]

Model answer (109 words; practice model, not official): 1. NDMA is the Prime Minister-chaired apex authority that lays down policies, plans and guidelines and exercises statutory functions under the Act. 2. The National Executive Committee is chaired by the Union Home Secretary and coordinates, monitors and implements national disaster-management policy and plans. 3. The State Disaster Management Authority is chaired by the Chief Minister and lays down state policy and approves the State Plan. 4. The State Executive Committee is chaired by the Chief Secretary and coordinates and monitors state-level implementation. 5. The District Disaster Management Authority is chaired by the District Collector or Magistrate, with the elected local-authority representative as Co-Chairperson, and prepares the District Plan.

Q10. Distinguish inflation, disinflation, deflation, reflation and stagflation.

[10 marks | 150 words]

Model answer (135 words; practice model, not official): Inflation is a sustained rise in the general price level; its rate is the percentage change in a stated index. Disinflation means this positive rate falls, so prices may still rise. Deflation is a sustained fall in the general price level and can increase real debt burdens and weaken output. Reflation is deliberate policy support to restore demand and prices from depressed conditions. Stagflation combines inflation with stagnant output and weak employment, often after an adverse supply shock. India's pandemic experience shows why the distinctions matter: logistics disruption raised essential prices despite slack. Demand-led inflation supports restraint; supply-led stagflation requires bottleneck repair plus credible expectation management. A single commodity spike is not automatically general inflation, while lower inflation does not restore the old price level. Precise vocabulary is therefore the first step in policy assignment.

Q11. How do clearing corporations, depositories and settlement design support market stability? Discuss their residual risks.

[15 marks | 250 words]

Model answer (225 words; practice model, not official): A securities trade is not complete when buyer and seller agree. Clearing validates obligations, netting reduces gross payment needs, settlement exchanges securities and funds, and custody records continuing ownership. Market infrastructure turns this chain into enforceable finality. A central counterparty, such as CCIL in covered Government-security and repo segments, novates trades and becomes buyer to every seller and seller to every buyer. Margins, collateral, default funds and loss-allocation rules reduce bilateral counterparty uncertainty. Delivery versus Payment links the securities and cash legs, limiting principal risk. Depositories hold securities electronically; under Depositories Act section 10, the depository is registered owner only for transfer, while the investor remains beneficial owner with substantive rights and liabilities. Shorter settlement reduces replacement exposure and frees collateral sooner. However, speed also compresses the time available for funding, securities delivery, corrections and fraud checks. Central clearing mutualises and manages risk but concentrates dependence on one critical institution. Netting lowers liquidity demand in normal conditions yet a member default can create sudden margin calls. Cyber failure, operational outage, wrong client records and inadequate recovery resources remain possible. Stability therefore requires more than a fast cycle. Regulators need robust participation standards, transparent margin models, stress testing, segregated client assets, default waterfalls, interoperable records, cyber resilience and tested recovery plans. The correct conclusion is that infrastructure transforms and contains risk; it does not abolish it.

Q12. Explain how buffer stocks stabilise food prices and why excess stocks create problems.

[15 marks | 250 words]

Model answer (192 words; practice model, not official): Buffer stocks combine operational inventory for routine NFSA and welfare distribution with strategic reserves for shocks. Quarterly norms reflect seasonal procurement and distribution: the minimum central-pool requirement is highest after major arrivals and lower as grain is issued. Stocks stabilise prices through two directions. Procurement absorbs eligible surplus when market prices weaken, supporting producers and building reserves. During shortage or inflation, NFSA releases protect entitled households and OMSS sales add market supply. Inter-State movement also transfers grain from procurement regions to deficit regions. The buffer therefore supports availability, access and stability. However, stock above a justified norm carries interest, storage, handling, preservation and quality costs. Slow rotation increases deterioration risk and occupies capacity needed for new arrivals. Large public inventories can crowd private storage, intensify fiscal subsidy and reinforce paddy-wheat procurement even in water-stressed regions. A stock labelled 'excess' may also be poorly located or already committed, so simple subtraction from the norm is insufficient. Reform requires dynamic stocking based on dated inventory, pipeline obligations and shock risk; modern silos and quality monitoring; transparent first-in-first-out rotation; decentralised procurement where efficient; and predictable OMSS triggers. The objective is adequate insurance, not maximum stock.

Q13. Describe the benefits of deriving electric energy from sunlight in contrast to the conventional energy generation. What are the initiatives offered by our Government for this purpose?

[15 marks | 250 words]

Model answer (143 words; practice model, not official): Solar electricity has no fuel purchase or combustion at generation, is modular and quick to deploy, and can serve utility, rooftop, pump and remote loads. It reduces marginal fuel-import exposure and local air pollution compared with fossil generation. Government instruments include solar parks and competitive procurement; PM Surya Ghar for rooftops; PM-KUSUM for farm applications; Green Energy Corridors; renewable obligations and RECs; domestic-manufacturing support; and storage procurement/VGF. Economic Survey 2025-26 reports 135.81 GW installed solar capacity by December 2025 and 8 GW rooftop capacity under PM Surya Ghar, but installed MW is not annual generation. Solar output is daytime-variable, needs land and transmission, and creates evening ramps, mineral and recycling issues. Solar's advantage is therefore strongest when capacity is integrated with grids, storage, demand response, domestic capability and ecological siting, and judged by reliable delivered energy rather than the winning bid.

Q14. Design a balanced manufacturing strategy for India.

[15 marks | 250 words]

Model answer (201 words; practice model, not official): India's manufacturing strategy must combine horizontal foundations with selective capability building. Reliable power, logistics, serviced land, skills, finance, contract enforcement and predictable regulation lower costs for every firm. Targeted instruments are justified where learning, coordination or strategic-dependence failures are demonstrable. Economic Survey 2025-26 identifies five reform pillars: ease of doing business, R&D and innovation, skilling, infrastructure and logistics, and scaling MSMEs. Industrial clusters can pool suppliers, laboratories, workers and transport, but a park becomes an ecosystem only through utilisation and institutional agility. PLI can help anchor scale, while Udyam, CGTMSE, TReDS, ZED, cluster facilities and procurement help smaller suppliers meet quality, cost and delivery requirements. Make in India mobilises investment; Atmanirbhar Bharat should mean resilient, competitive capability rather than autarky. Policy must protect competition. Large anchor firms can organise value chains, yet repeated support may entrench incumbents. Imports of technology and inputs can accelerate learning, while indiscriminate localisation may raise cost. Success should be measured through additional investment, domestic value, exports, productive jobs, supplier spillovers, innovation, regional distribution and fiscal cost. States should compete on implementation quality, while national standards preserve a common market and limit subsidy races. Periodic review and sunset convert industrial policy from discretion into disciplined experimentation.

Q15. Examine GST compensation as a problem of transition design and fiscal-federal trust.

[15 marks | 250 words]

Model answer (197 words; practice model, not official): The GST (Compensation to States) Act, 2017 created transition insurance, not ordinary devolution. It protected defined State revenue for five years from GST rollout, using 2015-16 as the base and 14 per cent annual protected growth. A cess on specified luxury and demerit goods financed the fund, reducing the perceived risk of surrendering State indirect-tax bases. COVID-19 exposed the design's financing weakness. Collections fell while protected claims continued, creating a gap. Back-to-back Union borrowing transferred resources to States, and cess collection continued beyond the original period to service that liability. The compensation entitlement ended in June 2022; loan servicing and cess continuation were distinct. Exit also required date discipline. The GST Council Secretariat recorded that Notification 03/2025-Compensation Cess (Rate), dated 31 December 2025, made specified pan-masala and tobacco entries nil from 1 February 2026 alongside revised GST treatment. The episode yields three lessons: federal reforms need credible transition insurance; shock-financing responsibilities should be specified beforehand; and recommendations, statutory entitlement, borrowing and notified levy must remain distinct. Permanent compensation could weaken reform incentives, but abrupt withdrawal can damage trust. Future transitions should combine transparent baselines, independent reconciliation, time-bound support and a pre-announced exit linked to measurable liabilities.

Q16. Define ecosystem carrying capacity and explain its relevance to sustainable planning.

[15 marks | 250 words]

Model answer (140 words; practice model, not official): 1. available evidence indicates that an ecosystem as interacting biotic communities and their abiotic physical and chemical environment exchanging energy and matter; A. G. Tansley coined the term in 1935, and every claim must state the practical system boundary rather than treat all nature as one unit. 2. Stability or resistance describes limited change under disturbance, whereas resilience describes recovery after disturbance; a uniform plantation can appear stable yet remain poorly resilient to a specific shock. 3. Carrying capacity is the population or project load an ecosystem can sustain over time without eroding regenerative functions; it is a conditional threshold shaped by consumption, technology and governance, not a fixed headcount. 4. Forest or tree-cover extent is a canopy measure and cannot by itself establish native composition, age structure, biodiversity, resilience or ecosystem functioning; extent and ecological quality are different claims.

Q17. Analyse the Gaganyaan qualification ladder before crewed flight.

[15 marks | 250 words]

Model answer (226 words; practice model, not official): 1. Human rating is a system-level qualification of launcher margins, redundancy, health monitoring, failure response and abort capability; an ordinary launch record does not establish crew safety. 2. Gaganyaan's Orbital Module combines a habitable, re-entry-capable Crew Module with a Service Module that supplies propulsion, power and other support; neither module alone is the complete flight element. 3. The Crew Escape System is designed to pull the Crew Module away from a failing launcher during relevant ascent phases; an abort demonstration validates escape logic, not a full orbital mission. 4. Environmental Control and Life Support maintains atmosphere, pressure, temperature, humidity and carbon-dioxide removal; it is a defining crewed-spacecraft system rather than a satellite payload. 5. Component and ground tests, air-drop recovery tests, test-vehicle abort demonstrations, uncrewed orbital flights and crewed flights are separate qualification rungs that answer different safety questions. 6. An Integrated Air Drop Test or main-parachute air-drop test uses a module simulator released from an aircraft or helicopter to qualify deceleration and recovery; no orbital launch vehicle is involved. 7. A Gaganyaan uncrewed orbital precursor tests integrated flight systems without accepting crew risk, while a crewed mission is a later and more demanding rung; a target date is not an achieved flight. 8. A crewed mission requires guided re-entry, parachute deployment, splashdown location, crew extraction and medical support; launch success alone cannot certify safe return.

Q18. Examine India's cooperative-federal counter-terror architecture with reference to the NCTC controversy.

[15 marks | 250 words]

Model answer (135 words; practice model, not official): 1. After the 2008 Mumbai attacks, NIA was created, MAC was strengthened, NATGRID was proposed, NCTC was debated, NSG hubs were added and coastal-security arrangements were revamped; these bodies have different functions. 2. The NIA Act, 2008 creates a central investigating agency for scheduled offences with concurrent jurisdiction; NIA does not replace the local police's first response, evidence preservation or ordinary policing role. 3. MAC and SMAC are intelligence-sharing and coordination platforms, not arresting, investigating or prosecuting commands. 4. The proposed NCTC faced objection because independent search, arrest and investigation powers under an intelligence body could bypass State policing competence; the coordination need did not automatically justify centralisation. 5. Local police ordinarily reach an attack first and secure the scene, witnesses, evidence and public order, making State capacity structurally irreplaceable even in nationally investigated cases.

Q19. Explain structural and non-structural measures for earthquake-risk reduction.

[15 marks | 250 words]

Model answer (155 words; practice model, not official): 1. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. 2. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. 3. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. 4. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. 5. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. 6. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone.

Q20. Evaluate the investor-protection architecture of mutual funds in India.

[15 marks | 250 words]

Model answer (202 words; practice model, not official): A mutual fund separates functions across a trust, asset management company, trustees and custodian. Investors hold units representing a proportionate claim on scheme net assets; they do not directly own each security. This architecture limits commingling and assigns portfolio management, oversight and custody to different entities. SEBI's Mutual Funds Regulations dated 16 January 2026, effective 1 April 2026, and Master Circular dated 20 March 2026 provide the current framework. Scheme Information Documents, Key Information Memorandums, portfolio and expense disclosures, valuation rules and the Riskometer support informed choice. Direct and regular plans disclose distribution-cost differences. The Riskometer's six levels communicate portfolio risk but neither predict return nor cap loss. Liquidity safeguards matter because open-ended redemption can conflict with illiquid assets. Stress testing, borrowing limits, fair valuation and segregated portfolios after specified credit events help allocate losses equitably. Trustees oversee compliance, while custodians safeguard assets. Limits remain. Disclosure can overwhelm investors, credit ratings can change late, NAV can lag executable prices in stress and side pockets isolate rather than erase loss. Distributor incentives can create mis-selling. Protection must therefore combine understandable labels, cost transparency, liquidity management, independent oversight, suitability and prompt remedy. Regulation should enable diversified household participation without implying guaranteed capital or performance.

END OF DETAILED SOLUTIONS